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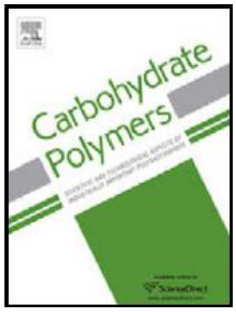

Title: Comparative study of physicochemical properties and bioactivity of *Hericium erinaceus* polysaccharides at different solvent extractions

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Comparative study of physicochemical properties and bioactivity of *Hericium erinaceus* polysaccharides at different solvent extractions

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Highlights

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- Four different solvents were used to extract *Hericium erinaceus* polysaccharides (HEPs).
- Extraction solvents had significant effects on physicochemical properties of HEPs.
- HEP-C extracted with citric acid showed stronger bioactivities than HEP-W and HEP-S.
- Citric acid can be used in food processing and for the extraction of bioactive PSs.

Abstract

In this study, hot water, 0.9% NaCl, citric acid, and 1.25 M NaOH/0.05% $NaBH_4$ were separately used for the extraction of water-soluble *H. erinaceus* polysaccharides (HEPs; HEP-W, HEP-S, HEP-C, and HEP-A) from the fruit body of *Hericium erinaceus*. The physicochemical properties and biological activities were then investigated and compared. Results showed that the extraction solvents exhibited significant effects on the extraction yields, molecular weights, monosaccharide compositions, preliminary structural characteristics, microstructures of HEPs and on their contents, such as neutral sugar, uronic acid, protein, and β-(1→3)-glucan. In vitro antioxidant activity assays indicated that HEP-C extracted with citric acid solution showed stronger scavenging abilities on hydroxyl and DPPH radicals and antioxidant capacities than HEP-W and HEP-S. Moreover, HEP-C exhibited the strongest inhibitory effects on α-glycosidase and α-amylase activities. Therefore, HEP-C extracted with citric acid can be developed as a potential bioactive ingredient for applications in food, medicine, and cosmetics industries.

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Keywords: *Hericium erinaceus* fruit body; Water-soluble polysaccharides; Solvent extractions; Physicochemical properties; *In vitro* antioxidant activity

Chemical compounds studied in this article:

Sodium chloride (PubChem CID: 5234)

Sodium hydroxide (PubChem CID: 14798)

Sodium borohydride (PubChem CID: 22959485)

N-butyl alcohol (PubChem CID: 263)

Chloroform (PubChem CID: 6212)

Trifluoroacetic acid (PubChem CID: 6422)

Pyridine (PubChem CID: 1049)

Acetic acid (PubChem CID: 176)

Ethanol (PubChem CID: 702)

Acetic anhydride (PubChem CID: 7918)

1. Introduction

Hericium erinaceus (Bull.) Pers., which is generally called Houtou or Shishigashira in China and Yamabushitake in Japan, is an edible mushroom that can also be used in medicinal practices belonging to the Aphyllophorales, Hydnaecase, and Hericium families (Friedman, 2015; Thongbai et al., 2015). *H. erinaceus* has been commonly prescribed in traditional Chinese medicine of its benefits to human health. It has been used in traditional folk medicine and consumed as a tonic food in China, Korea, and Japan (Wang et al., 2014). Over the past

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decade, *H. erinaceus* has attracted increasing attention in the fields of functional foods and biomedicine. *H. erinaceus* exhibits a broad spectrum of physiological and health-promoting functions, such as anti-aging and anticancer functions and functions against gastritis and metabolic diseases (Khan, Tania, Liu, Rahman, 2013; Wang et al., 2014). Modern pharmacological studies have demonstrated that *H. erinaceus* polysaccharides (HEPs) represent a major class of bioactive ingredients of *H. erinaceus* and have multiple health benefits, including immunomodulatory (Shen et al., 2017), antitumor (Kim et al., 2011), antigastritic (Wang, Gao, Xu, Gao, 2015), antioxidative and hepatoprotective (Han, Ye, Wang, 2013; Zhang et al., 2012), antihyperlipidemic (Shang et al., 2015), and antihyperglycemic activities (He et al., 2017). Therefore, the specific structures and sugar compositions of HEPs are potentially useful for the dietary application of *H. erinaceus* as functional foods.

In the recent years, a series of bioactive HEPs with different molecular weights and chemical structures have been extracted and isolated from the mushroom fruit bodies, mycelia, and fermentation broths of natural and cultured *H. erinaceus* (Friedman, 2015; He et al., 2017). As the first and most important step, extraction plays an important role in the characterization and utilization of bioactive HEPs from *H. erinaceus*. HEPs existed as structural components of the fungal cell wall, and the selection of suitable extraction methods depends on the cell wall structure (Zhang et al., 2011). In general, water extraction has been widely used as a classic method for the extraction and preparation of HEPs and other polysaccharides (PS) in the laboratory and industry application because of its dominant advantages, such as simplicity in application, low cost, safety, and environment-friendly properties (Nie & Xie, 2011; Yang, Zhao, Lv, 2008). High temperature can accelerate the

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dissolution of PS from the cell walls. Meanwhile, saline, acidic, and diluted alkali solutions are often used for PS extraction. These solutions break the cell walls from the outer layer to the inner layer under mild and strong conditions (Nie, Zhang, Li, Xie, 2013). For example, Gottschalk et al. (1994) prepared a capsular PS and long-chain lipopolysaccharides of *Actinobacillus pleuropneumoniae* serotype 1 as antigens for the serodiagnosis of swine pleuropneumonia by using a saline boiled extract. Zhang's group established a new and environment-friendly isolation method to obtain lentinan from the fruiting bodies of *Lentinus edodes* through alkali extraction with 1.25 M NaOH/0.05% $NaBH_4$ solution. Their results showed that the isolated lentinan had a yield of up to 5% and had a triple-helix conformation (Wang, Xu, Zhang, 2008; Zhang, Zhang, Cheng, 1999). Imbs et al. (2015) obtained the fucose-containing sulfated PS with excellent antioxidant activity from *Fucus evanescens* through hot acidic extraction and extraction with aqueous calcium chloride solution. Peasura et al. (2015) extracted sulfated PS from *Ulva intestinalis* by using distilled water, 0.1 M HCl, and 0.1 M NaCl at 80 °C and different extraction times. They found that solvent extraction effectively enhances antioxidant activity by a distinct molecular weight and chemical characteristic of the sulfated PS. Yao et al. (2017) compared hypolipidemic and antioxidant capacities of PS obtained from *Laminaria japonica* by using different extraction media (water, citric acid, sulfuric acid, hydrochloric acid, and phosphoric acid), and their results showed that extraction with water acidified with citric acid increases PS yield and bile salt adsorption rate and enhances free radical scavenging ability and antioxidant capacity. Thus, solvent extraction has a significant effect on PS yield and on the physicochemical properties, and structural characteristics, biological activities, and functionalities of PSs. However, to the best

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of our knowledge, no study or only a few study has reported the comprehensive evaluation of different solvent extractions on the physicochemical properties and bioactivities of HEPs from *H. erinaceus*.

Therefore, in the present study, we aimed to extract water-soluble HEPs from the fruiting body of *H. erinaceus* by using different extraction solvents (water, saline, acid, and alkaline solvents). The effects of extraction solvents on the physicochemical properties of HEPs were characterized using the total carbohydrate, protein, uronic acid, and β-1,3-glucans contents, as well as the monosaccharide composition, molecular weight, and preliminary structural features. The antioxidant and hyperglycemic activities of these HEPs in vitro were also evaluated and compared.

2. Materials and methods

2.1. Materials and Chemicals

The fruiting body of *H. erinaceus* was obtained from Shanghai Guosen Biotechnology Co., Ltd. (Shanghai, China). The strain used was H0605, which was obtained from the Herbarium of Edible Fungi Culture Collection Center Branch of the China Culture Collection of Agricultural Microorganisms. The fruit body (500 g) was ground into powder, passed through a 60 mesh sieve, and then treated with refluxing petroleum ether twice for 6 h for the removal of lipids and pigments. The residue was air dried and sealed in airtight plastic bags before use. Monosaccharide standards [D-Arabinose (D-Ara), D-glucose (D-Glu), D-galactose (D-Gal), D-mannose (D-Man), L-rhamnose (L-Rha), and D-xylose (D-Xyl)], glucuronic acid, gallic acid, trifluoroacetic acid (TFA), curdlan, acarbose, porcine pancreatic α-amylase,

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α-glucosidase, p-nitrophenyl-α-D-glucopyranoside (pNPG), hydrogen peroxide (H_2O_2), 1,1-diphenyl-2-picrylhydrazyl (DPPH), 2,2´-azinobis (3-ethylbenzothiazoline-6-sulphonic acid) (ABTS), 6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (Trolox) and 2, 4, 6-tris(2-pyridyl)-s-triazine were purchased from Sigma-Aldrich Chemical Co. (St. Louis, MO, USA). Sodium chloride, sodium hydroxide, sodium borohydride, potassium persulphate, ferrous sulfate, aniline blue, citric acid, methanol, ethanol, chloroform, *n*-butyl alcohol, acetic acid, petroleum ether, acetic anhydride, pyridine, sulfuric acid, carbazole, hydroxylammonium chloride and phenol were of analytical grade (AR) purchased from local suppliers.

2.2. Polysaccharide extraction and isolation

We separately used hot water, 0.9% NaCl, citric acid (pH 3.0), and 1.25 M NaOH/0.05% $NaBH_4$ solutions to extract HEPs from the fruiting body of *H. erinaceus*. 50.0 g of pretreated powder was extracted twice with 500 mL of distilled water at 95 °C for 8 h, 500 mL of 0.9% NaCl aqueous solution at 95 °C for 8 h, 500 mL of citric acid aqueous solution (pH 3.0) at 95 °C for 8 h, or 500 mL of 1.25 M NaOH/0.05% $NaBH_4$ aqueous solution at 25 °C for 3 h. After centrifugation (DL-6C, Shanghai Anting Scientific Instrument Factory, China), the resulting supernatants (1000 mL) of the citric acid and 1.25 M NaOH/0.05% $NaBH_4$ solutions were neutralized by adding 0.5 M NaOH and 36% acetic acid, respectively. Afterward, the four aqueous extracts were concentrated to 300 mL using a rotary evaporator (RE-2000, Shanghai Yarong Biochemical Instrument Factory, China), respectively, and precipitated with 4 volumes of 95% ethanol at 4 °C for 12 h, followed by

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centrifugation at 5000 rpm for 20 min. Four precipitates were washed three times with absolute ethanol. Then, the washed precipitates were dissolved in distilled water, deproteinized with a Sevag reagent (Staub, 1956), dialyzed with 3500 Da molecular weight cutoff membrane against distilled water for 72 h, and freeze-dried (Alpha 2-4, Martin Christ Gefriertrocknungsanlagen GmbH, Osterode am Harz, Germany) for the acquisition of partially purified PSs, namely, HEP-W (4.05 g), HEP-S (4.83 g), HEP-C (4.20 g), and HEP-A (5.88 g). The detailed extraction and isolation procedures of HEPs are summarized in Fig. 1.

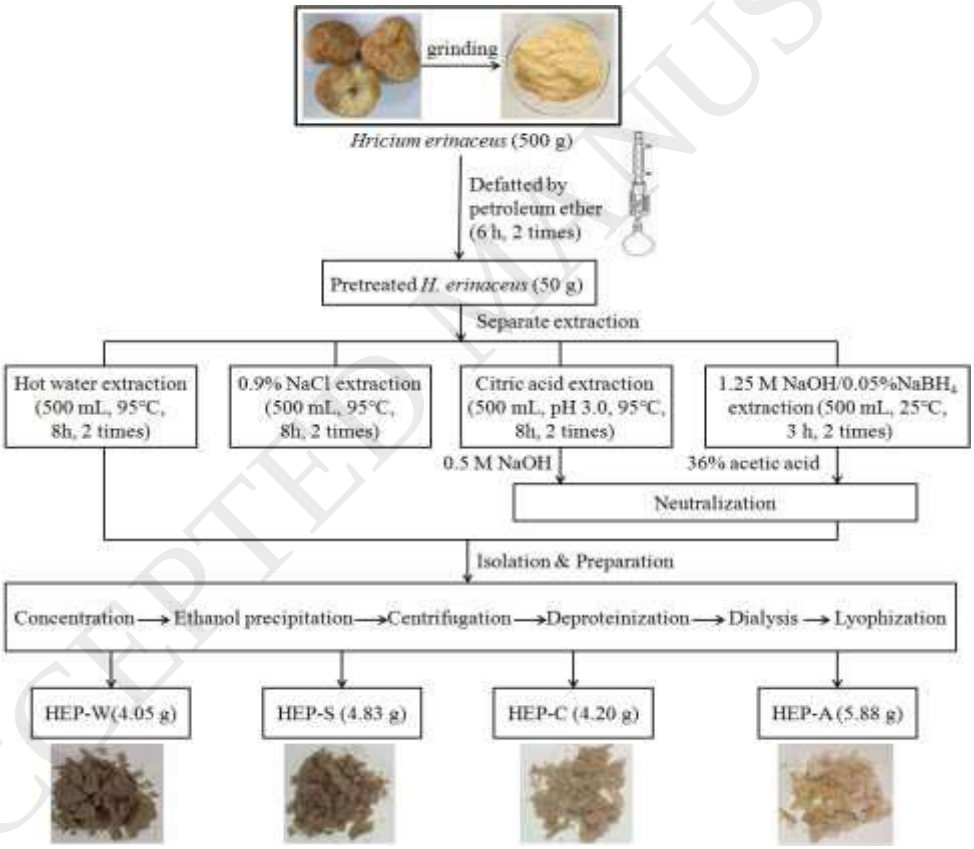

Fig. 1. Schematic routes for PS extraction and isolation of from the fruiting body of *H. erinaceus* by using different solvent extractions.

2.3. General physicochemical properties of HEPs

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The total carbohydrate, uronic acid, and protein contents of HEPs were determined by the phenol–sulfuric acid method using glucose as a standard (Dubois, Gilles, Hamilton, Rebers, & Smith, 1956), via sulfuric acid–carbazole method using glucuronic acid as a standard (Bitter, & Muir, 1962), and via the Bradford method using bovine serum albumin (BSA) as a standard (Bradford, 1976), respectively. Total phenolic content was determined by Folin-Ciocalteu assay reagent using gallic acid as a reference (Siu, Chen, Wu, 2014). The extraction yield (%) of HEPs was calculated by the following equation: extraction yield (%, w/w) = [weight of dried HEPs (g)/weight of raw materials (g)] × 100%. The intrinsic viscosity [η] of HEPs in water was determined at 25±0.1 °C with an Ubbelohde capillary viscometer (type ϕ 0.4-0.5 mm, Shanghai Shenyi Glass Instrument Factory, China). The kinetic energy correlation was kept negligible.

2.4. Determination of molecular weight and molecular weight distribution (MWD)

The weight–average molecular weight (M_w), number–average molecular weight (M_n), and MWD (M_w/M_n) of the HEPs were determined using high-performance size exclusion chromatography coupled to a multiangle laser light scattering (HPSEC–MALLS, DAWN HELLOS II λ=658 nm; Wyatt Technologies Corporation, USA). HPSEC–MALLS was performed on an Agilent 1260 HPLC system (Agilent, USA) equipped with an Agilent G1362A refractive index detector (RID, Agilent) and two SEC columns (OHpak SB-806 M HQ and SB-805 HQ, 8 mm Φ×30 cm, Shodex, Japan) in a series at 25 °C. A total of 0.1 M NaCl was used as the elution solvent at a flow rate of 0.5 mL/min. Moreover, HEPs were dissolved in this solvent at 1.0 mg/mL and filtered through 0.45 μm membranes (Millipore,

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USA) prior to injection. The *dn*/*dc* value used was 0.125. The on-line ASTRA 6.1 software package (Wyatt Technologies, USA) was used for data collection and analysis.

2.5. Monosaccharide composition analysis

A total of 10 mg freeze-dried HEP samples were hydrolyzed with 5.0 mL of 2 M TFA at 100 °C for 8 h in a sealed tested tube. After the samples were evaporated dry under reduced pressure, the resulting hydrolysate was treated with 10 mg $HONH_2 \cdot HCl$ and 1.0 mL of pyridine at 90 °C for 30 min and then treated with 1.0 mL of acetic anhydride at 90 °C for 30 min. The acetylated derivatives were filtered through 0.45 μm membranes (Millipore, USA) and analyzed by gas chromatography (GC) on an Agilent 7890A instrument (HP-5 fused-silica capillary column, 30 m×0.25 mm×1 μm) and a flame ionization detector (FID). The column temperature was held at 180 °C for 3 min, then increased to 280 °C at 10 °C/min, and held for 2 min. The monosaccharide standards were also derived and measured according to the same procedure as above.

2.6. UV–Vis and FT–IR spectroscopic analyses

UV–Vis spectra of the HEPs (1.0 mg/mL) were recorded in a spectrophotometer (Carry-100, Varian, USA) in the wavelength range of 190–400 nm at room temperature. Fourier transform–infrared (FT-IR) spectra of HEPs were determined using a Nexus 670 FT–IR spectrometer (Thermo Nicolet Co., USA) in the wavenumber range of 500–4000 cm^{-1} with KBr pellets referenced against air.

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2.7. β-1,3-glucan content determination

The total β-1,3-glucans contents of HEPs were determined using the reported methods (Ko & Lin, 2004; Nitschke et al., 2011). The working solution was previously prepared by mixing 127 mL of distilled water, 37 mL of glycine/NaOH buffer (1.0 M glycine, 1.25 M NaOH), 13.1 mL of 2 M HCl, and 25 mL of dye solution (5.0 g/L aniline blue). Afterward, we stirred the solution evenly and then stored it in the dark for 12 h to decolorize it from blue to yellow. Meanwhile, the HEP samples were predissolved in distilled water and kept at 4 °C for 12 h for complete hydration. A total of 0.4 mL of the sample and 3.6 mL of the working solution were mixed into a 5 mL centrifuged tube and incubated at 50 °C for 30 min. After the tubes were cooled to 25 °C, the fluorescence intensity was measured using a Cary Eclipse fluorescence spectrophotometer (Agilent, USA). The fluorescence was measured at an emission wavelength of 502 nm, excitation wavelength of 398 nm, and spectral bandwidth of 20 nm. The total β-1,3-glucans contents were derived from the calibration curve obtained by curdlan in the 0–15 μg/mL concentration range (R^2=0.9989).

2.8. Scanning electronic microscope (SEM) observation

The freeze-dried HEPs were fixed onto a copper stub. After sputtering with a layer of gold, the SEM images were observed and recorded using a S4800 SEM (JEOL Ltd., Japan) under a high vacuum condition at an accelerating voltage of 15 kV and image magnification of 500 ×.

2.9. Antioxidant activity assays in vitro

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The antioxidant activities in vitro of HEP-W, HEP-S, HEP-C, and HEP-A were evaluated by calculating the hydroxyl radical scavenging activity, DPPH radical scavenging activity, Trolox equivalent antioxidant capacity (TEAC) assay, and ferric reducing ability of plasma (FRAP) assay. For all assays, the HEP samples were predissolved in distilled water and measured at different concentrations (0-3.0 mg/mL) against distilled water as the blank control and the vitamin C (Vc) as a positive antioxidant reference.

For the adapted hydroxyl radical method, a volume of 0.2 mL of the tested solution (sample or control) was added into an equal volume of an aqueous solution of 5 mM $FeSO_4$. Then, a volume of 0.2 mL of 1% (v/v) H_2O_2 was added into the mixture under continuous stirring. After stirring and incubation at room temperature for 60 min, the absorbance of the resulting solution was measured at 510 nm by a UV-1600 Spectrophotometer (Beijing Ruili Analytical Instrument Co., Ltd., China). The hydroxyl radical scavenging activity (%) was estimated by the following formula: scavenging activity (%) = $(1-A/A_0) \times 100\%$, where A, and A_0 are the absorbance values of the sample and blank control, respectively.

Concerning the DPPH scavenging assay, a volume of 2 mL of the solution was added into 2 mL 0.1 mM DPPH solution in methanol. The reaction mixture was stirred and incubated at 25 °C for 30 min in the dark. The absorbance of the resulting solution was measured at 517 nm using a UV-1600 Spectrophotometer. The DPPH radical scavenging activity was calculated based on the following equation: scavenging activity (%) = $[1-(A_1-A_2)/A_0]\times 100\%$, where A_0, A_1, and A_2 are the respective absorbance values of the blank control, the sample with DPPH solution, and the sample with distilled water.

As for TEAC assay, the TEAC of HEPs was tested against $ABTS^{\cdot+}$ radical, which was

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generated from the oxidation of ABTS by potassium persulphate. The TEAC values were derived from the calibration curve generated with Trolox (0-30 μM). In the FRAP assay, ferric to ferrous ion reduction at low pH led to the formation of a colored ferrous-tripyridyltriazine complex, and the FRAP values of the samples were obtained by calibration with ferrous sulphate (0-30 μM). Details of the operating conditions and methods were described as previously reported (Leung, Zhao, Ho, & Wu, 2009).

2.10. In vitro antihyperglycemic activity assays

The anti-hyperglycemic activity in vitro of HEPs was evaluated with α-glucosidase and α-amylase inhibitory activity assays. For the α-glucosidase inhibition test, pNPG, which can be decomposed by α-glucosidase for the release of *p*-nitrophenol and can be detected by a color reagent at 405 nm, was used as substrate. For the two assays, the HEP samples were previously dissolved in distilled water at various concentrations against distilled water as the blank and acarbose as positive control. The details of the operating conditions and methods were described in details before (Yan et al., 2017).

2.11. Statistical analysis

All experiments are conducted in three replicates and the mean ±standard deviation (SD) is used in the analysis. The statistical analysis was performed by Student's *t*-test and analysis of variance (ANOVA) using OriginPro Software Version 8.0 (OriginLab Corp., MA, USA). $P<0.05$ indicated statistically significant differences.

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3. Results and discussion

3.1. Extraction yields and physicochemical properties of HEPs

The extraction yields and physicochemical properties of the four water-soluble HEPs from the fruit body of *H. erinaceus* obtained through different extraction methods are summarized in Table 1. The extraction methods had significant influences on the extraction yields and physicochemical properties. Depending on the extraction solvent, the yield of HEPs varied (approximately 8.10%–11.76%). Among these HEPs, HEP-A had the highest yield (11.76%), followed by HEP-S (9.66%), HEP-C (8.84%), and HEP-W (8.10%). In general, insoluble PSs can be effectively released from the cell wall and converted into soluble PSs through the disruption of the hydrogen bonds between the cellulose and hemicellulose during the alkali extraction of the cell wall materials, which would help increase the final recovery (Lahaye & Robic, 2007). Moreover, the HEP yield was lower with citric acid than with saline and alkaline at the same extraction time and temperature because of partial acid hydrolysis. In the process of extraction, the citric acid solution, especially at 95 °C, can degrade the long-chain PSs into short-chain PSs to some extent; even free sugars may be not precipitated by ethanol, leading to the lower HEP yields obtained (Yuliarti et al., 2008). Moreover, the lowest extraction yield of HEPs was obtained during water extraction because water only dissolves long-chain soluble PS. This result was in agreement with those of Peasura et al. (2015), who reported lower yield with acid extraction than with alkali extraction.

The highest total sugar content was obtained from HEP-C (81.51%) via citric acid extraction, and the lowest was obtained from HEP-W (72.30%) via hot water extraction.

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These results may be due to the fact that the glycosidic bonds in the PS chain are easily destroyed by citric acid (Yuliarti et al., 2015). The total protein content in the four HEPs except for HEP-A was low (1.90%–2.81%) because of the reduction of the isolated proteins in these HEPs via solvent extraction. Specifically, HEP-A extracted with alkali solution had the highest protein content (8.31%), corresponding to the release of proteins by the breakage of hydrogen bridges in the alkaline solution. This result also revealed that HEP-A may be a polysaccharide-protein complex (Yuliarti et al., 2015), which was verified by the UV–Vis spectrum analysis (data not shown). Uronic acid content (glucoronic acid equivalent) was highest in HEP-S (16.96%), followed by HEP-C (14.99%), HEP-W (13.98%), and HEP-A (12.14%), which indicated that the four HEPs were all acidic. Additionally, the four HEPs had small quantities of polyphenol (0.54%–0.77%). On the basis of the findings above, we can conclude that these differences may be attributed to the selective type of extraction method and the use of extraction solvent. The results were in agreement with those of the previous studies (Peasura et al., 2015; Yuliarti et al., 2015).

Table 1. Extraction yields, chemical properties, and compositions of PSs from *H. erinaceus* by using different solvent extractions.

Sample	HEP-W	HEP-S	HEP-C	HEP-A
Yield (%)	8.10±0.23	9.66±0.81	8.84±0.14	11.76±0.31
Carbohydrate (wt%)	72.30±1.70	73.97±2.30	81.51±2.56	79.17±2.76
Protein (wt%)	2.81±0.18	1.90±0.04	1.97±0.32	8.31±0.13
Uronic acid (wt%)	13.98±1.51	16.96±2.83	14.99±1.04	12.14±1.47
Polyphenols (wt%)	0.77±0.02	0.62±0.02	0.64±0.03	0.54±0.02
β-1,3-glucans (mg/g fruit body) [a]	0.23±0.05	0.38±0.05	1.26±0.08	23.21±0.75
Sugar composition (molar ratios) (mol%)				

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L-Rhamnose	9.9	1.7	9.0	3.3
D-Arabinose	1.6	1.0	2.0	2.4
D-Mannose	1.0	1.0	1.0	1.0
D-Glucose	35.5	30.7	40.7	66.5
D-Galactose	6.1	4.6	7.5	9.5

[a] The total β-1,3-glucans content was determined by an aniline blue fluorescence assay.

In the current work, a fluorescence assay based on the aniline blue dye, which binds specifically to β-(1→3)-glucans, was used for the determination of the total β-(1→3)-glucan contents of the four HEPs at different solvent extractions. As shown in Table 1, the extraction medium had significant influence on the total β-(1→3)-glucan contents. HEP-A extracted with alkali solution showed the highest content (23.21 mg/g fruit body), followed by HEP-C (1.26 mg/g fruit body), HEP-S (0.38 mg/g fruit body), and HEP-W (0.23 mg/g fruit body). The β-(1→3)-glucan constitutes one of the structural macromolecules in the cell wall of mushrooms or fungus; thus, the alkali solution can readily cause the cell wall to swell, which can destroy the hydrogen bonds in the cell walls, leading to the increased release of β-(1→3)-glucans. Consequently, the alkali solution resulted in the highest content of β-(1→3)-glucan; similar results were also observed in the previous studies (Ko & Lin, 2004; Zhang, Xu & Zhang, 2005).

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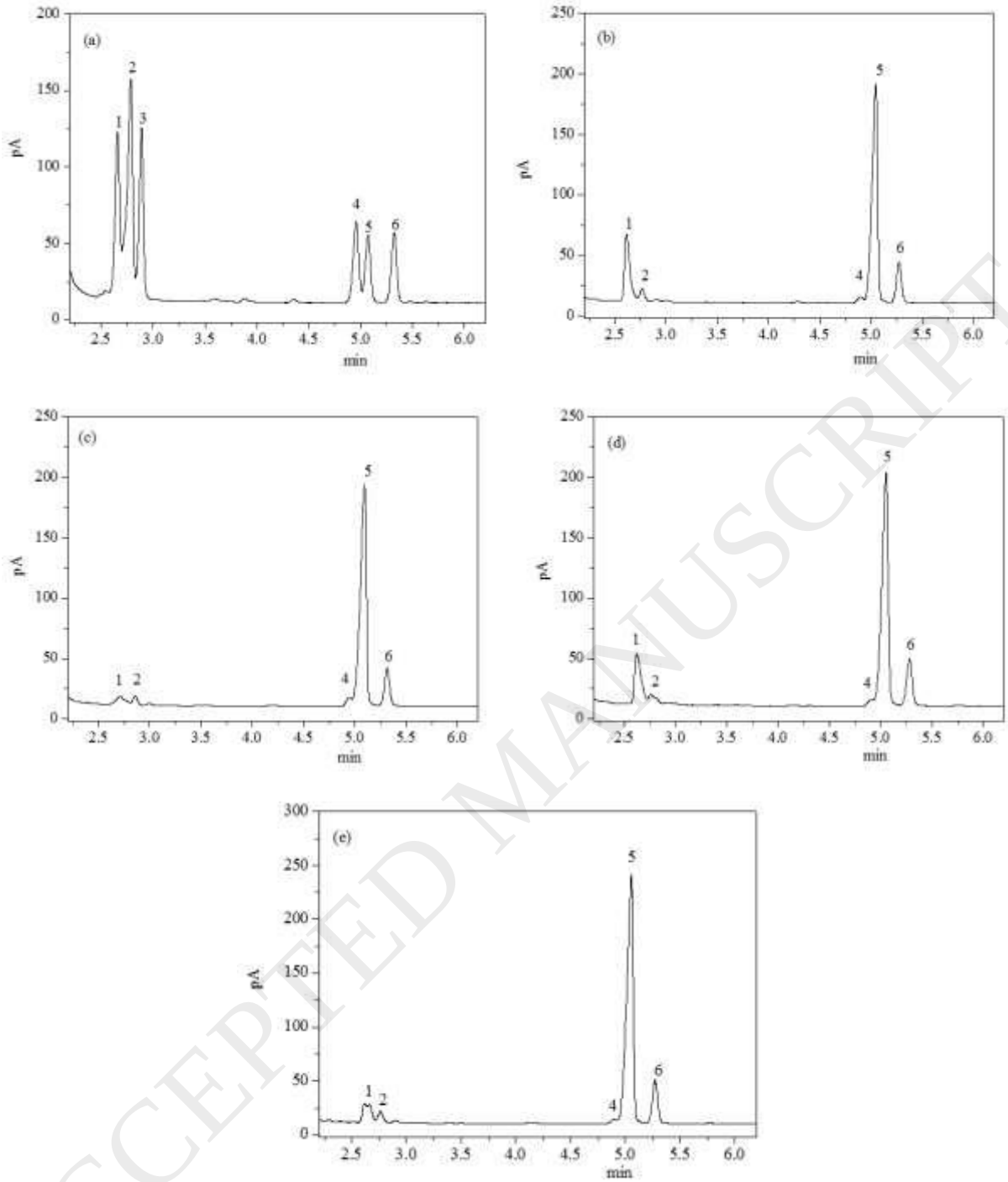

Fig. 2. GC profiles of nitrile acetates of (a) monosaccharide standards, (b) HEP-W, (c) HEP-S, (d) HEP-C, and (e) HEP-A. Peaks: 1, L-Rha (Rt: min); 2, D-Ara (Rt: min); 3, D-Xyl (Rt: min); 4, D-Man (Rt: min); 5, D-Glc (Rt: min); 6, D-Gal (Rt: min).

The four HEPs were completely hydrolyzed by TFA, converted into nitrile acetates, and

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then subjected to GC analysis; the obtained GC results are shown in Table 1 and Fig. 2. According to the monosaccharide standards, the four HEPs were all composed of L-Rha, D-Ara, D-Man, D-Glc, and D-Gal in different molar percentages. This finding suggested that the four HEPs extracted from *H. erinaceus* were heteropolysaccharides with different chemical compositions. Particularly, D-Glc was a predominantly monosaccharide in these HEPs. Zhang et al. (2012) reported that three fractions of the endopolysaccharides (HEP40, HEP60, and HEP80) from the mycelium of *H. erinaceus* all comprised glucose, galactose, mannose, arabinose, xylose, rhamnose, and fucose with varying molar ratios. Zhu et al. (2014) also found that the HEP monosaccharide compositions of HEPs extracted from the fruits of *H. erinaceus* by hot water are not considerably different from those extracted by enzyme-assisted methods, although they differ significantly with respect to molar ratio. Consequently, the variation in the monosaccharide composition of HEPs may be attributed to the differences in the source of raw materials and extraction methods and isolation procedures used.

In the current work, we performed HPSEC–MALLS–RID analysis to determine M_w, M_n, and MWD of the four HEPs, and the results are illustrated in Table 2 and Fig. 3. As shown in Fig. 3, two peaks were observed for each HEP in the HPSEC–MALLS curves regardless of the extraction solvent used. Differences in the elution profiles reflect differences in the molecular size. The results suggested that these HEPs possessed two MWDs that mainly consisted of high- and low-M_w fractions. HEP-W extracted with hot water presented two main peaks with the M_ws of 597.6 and 314.7 kDa, and the high fraction of ~72% was the dominant based on the peak area. Similarly, HEP-S obtained by saline extraction also showed

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two peaks with M_ws of 449.4 and 442.7 kDa, and the high M_w fraction of ~73% was the main composition. Alternatively, the M_ws of HEPs extracted with acid and alkali were significantly lower than those of HEPs obtained by hot water and saline. This result suggests that the PS chains may have been destroyed and degraded to some extent during the acid and alkaline extractions, then converted a portion of the high-M_w fractions into the low-M_w fractions, thereby increasing the amounts of low-M_w fractions in the HEP molecules. For example, the M_ws of HEP-C extracted with citric acid were 263.6 and 229.8 kDa with peak areas of 28% and 72%, respectively. The same results were also obtained in our previous studies (Wang, Pei, Ma, Cai, Yan, 2014; Yan et al., 2009).

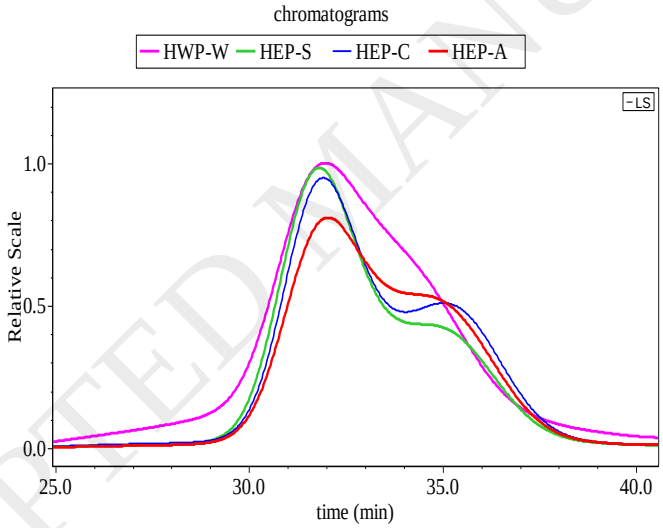

Fig. 3. HPSEC–MALLS curves of HEP-W, HEP-S, HEP-C, and HEP-A in 0.1 M NaCl aqueous solution at 25 °C.

As shown in Table 2, different extraction solvents had significant effects on the intrinsic viscosity ([η]) of the four HEPs at the same conditions. Specifically, HEP-C extracted with citric acid solution had the lowest [η] value (135.60 mL/g) than others possibly because of

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the acidic degradation of HEPs during the extraction. Thus, acidic conditions have significant effect on the distribution of molecular weights of polysaccharide (Shu & Lung, 2004), which is in agreement with the result of HPSEC–MALLS–RID analysis.

Table 2. Molecular characteristics of PSs from *H. erinaceus* by using different solvent extractions.

Sample	[η] (mL/g)	M_w (×10^3 Da) [a]	M_n (×10^3 Da) [b]	M_w/M_n [c]	%/Area
HEP-W	658.98	597.6	582.3	1.03	72.2
		314.7	204.4	1.54	27.8
HEP-S	620.24	449.4	413.2	1.09	73.6
		442.7	295.2	1.50	26.4
HEP-C	135.60	229.8	218.4	1.05	72.0
		263.6	162.1	1.63	28.0
HEP-A	242.27	256.8	249.7	1.03	67.9
		320.8	263.0	1.22	32.1

[a] Weight average molecular weight.

[b] Number average molecular weight.

[c] Molecular weight distribution.

3.2. Preliminary structural features and microstructures of HEPs

FT–IR spectroscopy is often used for the identification of characteristic organic groups in PSs. Fig. 4 displays the FT–IR spectra of HEP-W, HEP-S, HEP-C, and HEP-A. No obvious difference was observed among the characteristic organic groups of HEPs obtained from different extraction solvents. The result indicated that the different solvent extraction has no effect on sugar structure. The broad and strong absorption peak at approximately 3330

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cm^{-1} was assigned to the O–H stretching vibration, while the weak band at approximately 2921 cm^{-1} was attributed to the C–H stretching vibration. The two bands at approximately 1645 and 1417 cm^{-1} corresponded to the asymmetric and symmetric stretching of the carboxylate anions groups, further indicating that these HEPs are acidic (Manrique, & Lajolo, 2002). The strong extensive absorption peaks located at 1150, 1077, and 1030 cm^{-1} were ascribed to the stretching vibrations of the C–O–H side groups and the C–O–C glycosidic band vibration of a pyranose ring. These results further confirm that the four extracted HEPs contained pyranose sugar (Kacurakova et al., 2000). In addition, the characteristic absorption bands at approximately 890 and 908 cm^{-1} indicate that β-glycosidic bonds are the main bond type in these HEPs.

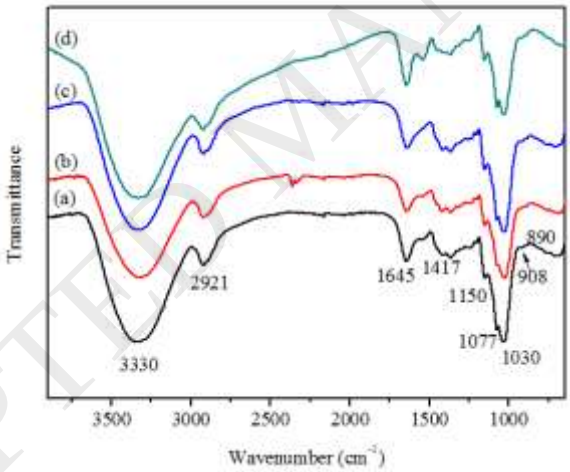

Fig. 4. FT–IR spectra of (a) HEP-W, (b) HEP-S, (c) HEP-C, and (d) HEP-A.

Fig. 5 displays the SEM images of the four HEPs (HEP-W, HEP-S, HEP-C, and HEP-A) at magnifications of 500×. As shown in Fig. 5, the surface morphologies of the four HEPs extracted using different solvents presented significant differences in shape and size. HEP-W extracted with hot water showed a massive and rough surface with many aggregates, forming

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large tracts of PS (Fig. 5a). Similarly, HEP-S extracted by saline solution presented clumpy surfaces with a small part of lumpish particles (Fig. 5b). However, HEP-C possessed relatively loose and flaky surfaces (Fig. 5c), which indicated that citric acid may have partially disrupted PS and decreased its size. This result was in line with those of HPSEC–MALLS–RID analysis. Compared with citric acid, the alkali solution had a more drastic effect on the cell walls of the fruiting bodies, and the disruption of the cells increased the contact area between the solid and liquid phases. This phenomenon leads to the relatively rough surface of HEP-A, with on a reticular layer and a nonuniform size (Fig. 5d). Thus, the SEM analysis verified that the alkali extraction produced the highest yield (Table 1).

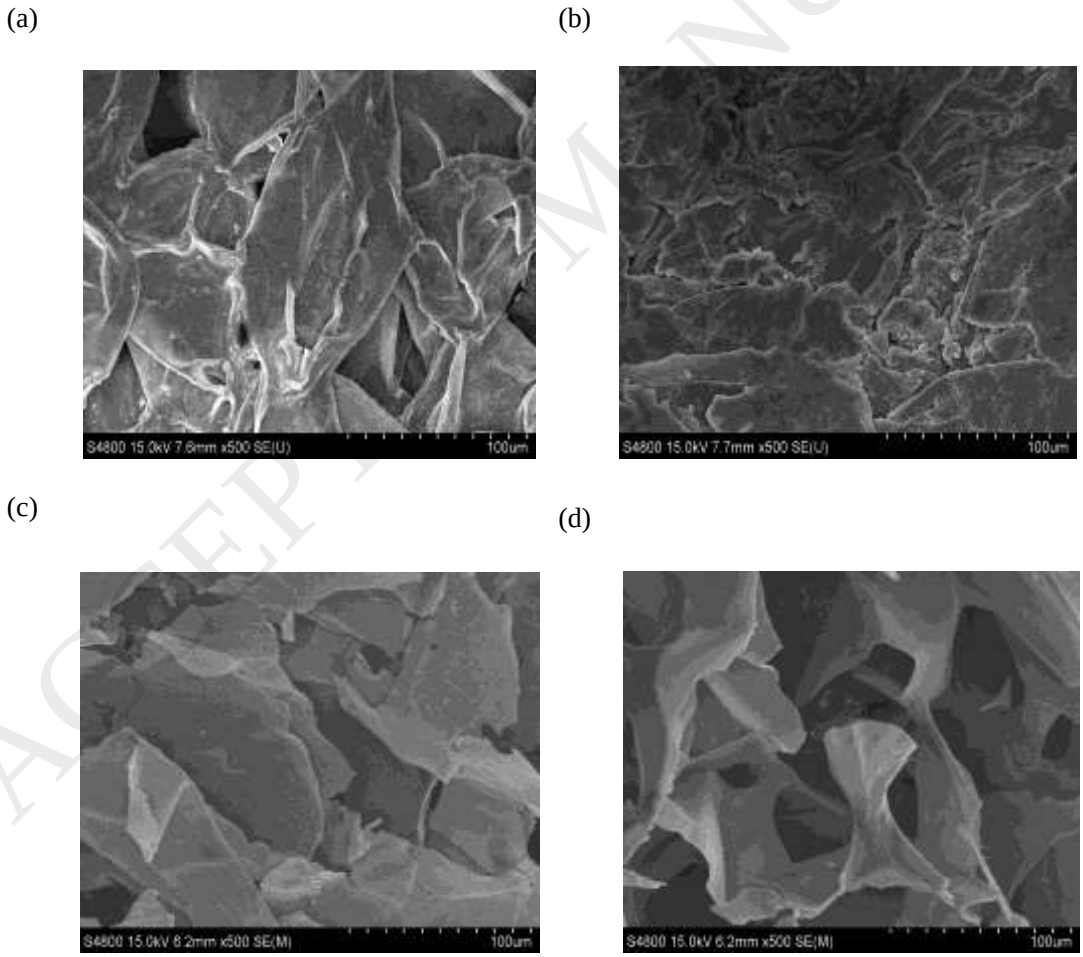

Fig. 5. Scanning electron micrographs (500 ×) of (a) HEP-W, (b) HEP-S, (c) HEP-C, and (d)

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HEP-A.

3.3. In vitro antioxidant activity of HEPs

The DPPH radical is a stable free radical that has been widely used for the estimation of the free-radical scavenging capacities of antioxidants (Matthaus, 2002). Fig. 6a shows the DPPH radical scavenging ability of the four HEPs (HEP-W, HEP-S, HEP-C, and HEP-A) extracted from *H. erinaceus* by using different extraction solvents. The four HEPs showed clear scavenging ability on the DPPH radical at all tested concentrations (0–3.0 mg/mL). At 3.0 mg/mL, the scavenging abilities of HEP-W, HEP-S, HEP-C, HEP-A, and Vc on DPPH radical were 62.31%, 62.42%, 67.97%, 68.26%, and 95.24%, respectively. Moreover, the IC_{50} values of HEP-W, HEP-S, HEP-C, HEP-A and Vc were calculated to be about 1.62, 1.55, 1.07, 1.29, 0.26 mg/mL, respectively. HEP-C and HEP-A showed the high DPPH scavenging but much weaker than that of Vc. The possible reason for this result may be the PS degradation by acidic and alkali treatments, which resulted in PS with low molecular weight (Table 2) and with enhanced hydrogen-donating ability. The DPPH radical scavenging ability of antioxidants may be related to the hydrogen-donating capacity. Notably, HEP-C with the lower molecular weight showed stronger a DPPH radical scavenging ability than HEP-A. Similarly, Zhang et al. (2013) reported that two PSs, that is, UFP and MFP, with decreased molecular weights extracted from *Flammulina velutipes* by ultrasonic and microwave treatments, respectively, showed higher DPPH radical scavenging abilities possibly because of the strong hydrogen-donating ability. Zhang et al. (2012) also prepared three fractions of the endopolysaccharides (HEP40, HEP60, and HEP80) from the mycelium of *H. erinaceus*

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grown on tofu whey by gradient concentrations of ethanol. The scavenging effects of HEP40, HEP60, and HEP80 (1.25 mg/mL) on DPPH radical were approximately 43%, 31%, and 60%, respectively, which was weaker than those of HEPs reported in this work. These results revealed that more bioactive PSs may be extracted using an acid or alkali solution than hot water, which was further verified by Wang et al. (2014).

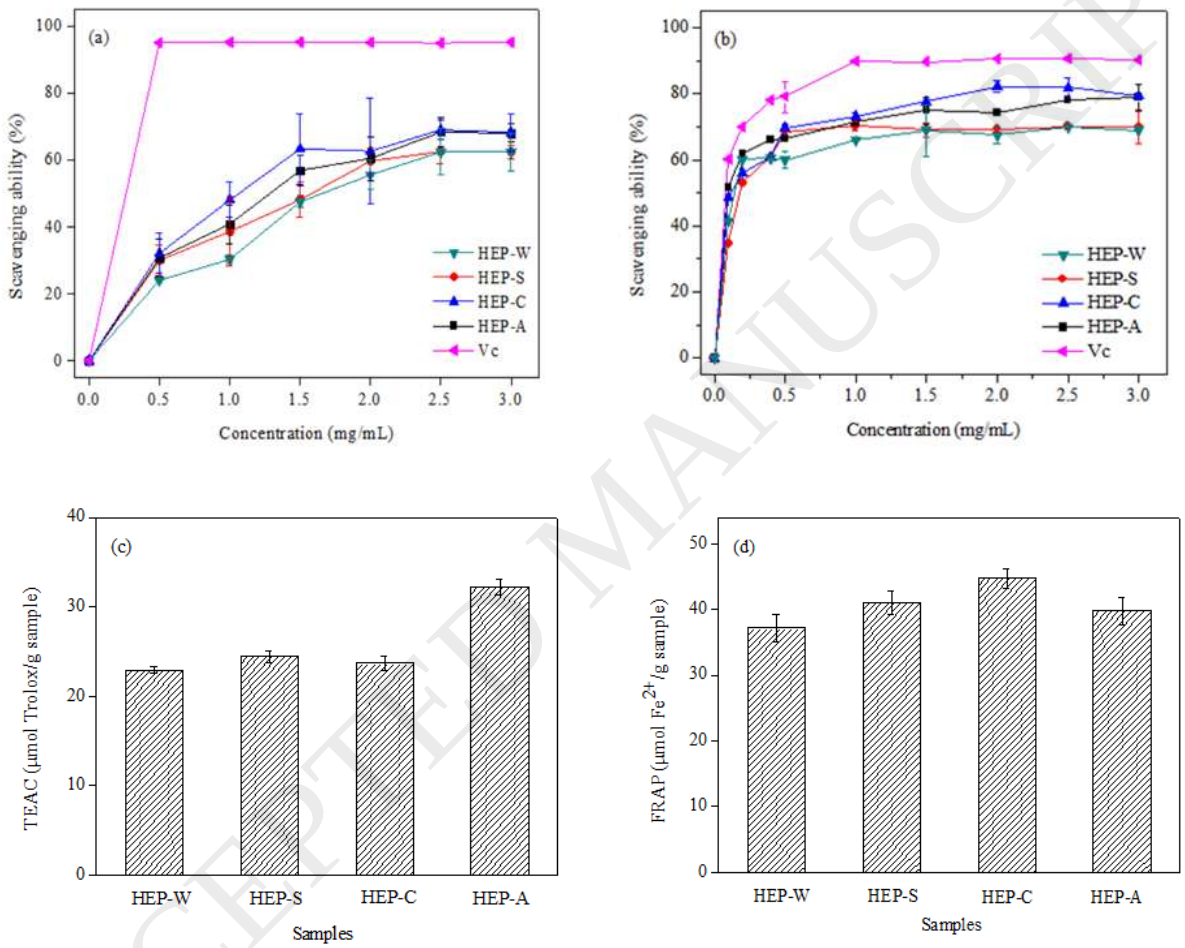

Fig. 6. Antioxidant activities in vitro of HPE-W, HEP-S, HPE-C, and HPE-A via the (a) scavenging ability on DPPH, (b) ·OH, (c) TEAC assay, and (d) FRAP assay. Each value is expressed as mean ± SD (n =3).

Hydroxyl radicals are typical reactive oxygen species generated in the body and can

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easily damage cells (Yuan, Zhang, Fan, Yang, 2008). Thus, the removal of hydroxyl radical is crucial to antioxidant defense in cells or food systems, and hydroxyl radical scavenging is extremely important in the evaluation of antioxidant activity in vitro. Fig. 6b presents the hydroxyl radical scavenging ability of the four HEPs using Vc as the positive control. The scavenging ability of these HEPs on the hydroxyl radical dose dependently increased with the increase in HEP concentration from 0 mg/mL to 3.0 mg/mL. At 3.0 mg/mL, the scavenging abilities of HEP-W, HEP-S, HEP-C, and HEP-A on the hydroxyl radical were 69.05%, 69.91%, 79.37%, and 79.11%, respectively, which were stronger than that of the result reported by Zhang et al. (2012), but less than that of Vc (90.43%). The results suggested that the four HEPs exhibited strong scavenging ability on the hydroxyl radical, and HEP-C extracted with citric acid showed the highest hydroxyl radical scavenging ability among these HEPs, which was consistent with the result of DPPH radical scavenging assay. Similar results were also found in the previous studies (Lu et al., 2013; Yao et al., 2017).

Figs. 6c and d show the in vitro antioxidant capacities of the four HEPs. The capacities were evaluated via the TEAC (on ABTS·+ radical scavenging activity) and FRAP assays (on ferric reducing activity). In the TEAC assay, HEP-A with a TEAC value of 32.2 μmol Trolox/g sample was obtained by alkali extraction. It had the highest antioxidant capacity among the tested HEP samples possibly because it had the highest protein content (8.31% vs. 1.90%–2.81%; Table 1). In the FRAP assay, HEP-C extracted with citric acid solution significantly showed the highest antioxidant capacity with a FRAP value of 44.82 μmol Fe^{2+}/g sample. These results suggested that extraction medium affected the antioxidant activities of HEPs. HEPs extracted from *H. erinaceus* through acid or alkali extraction

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possessed higher sugar and protein contents (Table 1) and lower intrinsic viscosities and molecular weights (Table 2) than those extracted through other methods, thereby exhibiting enhanced antioxidant capacities. This result was in line with those of DPPH and hydroxyl radical scavenging activities. Additionally, the total β-(1→3)-glucan contents had certain effects on the antioxidant activities of the HEPs.

3.4. Inhibitory effects of HEPs on α-glycosidase and α-amylase

In individuals with type 2 diabetes, α-glycosidase and α-amylase can hydrolyze starch to glucose during the PS catabolism, leading to a significant increase in the postprandial blood glucose (Omojokun, 2015). Thus, α-glycosidase and α-amylase inhibition assays can be regarded as simple and effective strategies for the evaluation of hypoglycemic activities of bioactive compounds in vitro (Lordan et al., 2013). Fig. 7 shows the inhibitory effects of the four HEPs on the α-glycosidase and α-amylase activities by using acarbose as the positive reference. The four HEPs had similar inhibitory effects on α-glycosidase and α-amylase activities, and they were dose-dependently increased as the HEP concentration was increased from 0 mg/mL to 5.0 mg/mL. HEP-C extracted with citric acid solution exhibited the strongest inhibitory effects on the α-glycosidase and α-amylase activities at all tested concentrations, whereas were weaker than that of acarbose. At 5.0 mg/mL, the inhibitory abilities of HEP-C on α-glycosidase and α-amylase were 73.66% and 65.92%, respectively, which suggested that HEP-C possessed stronger inhibitory ability against α-glycosidase than that of α-amylase under the same concentrations. This variation may be attributed to the different mechanisms of action between α-glycosidase and α-amylase, and a moderated inhibition of α-amylase and a stronger α-glycosidase inhibition should be preferred (Cho,

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Han, & You, 2011). Similarly, Zhao et al. (2017) reported that the PS from *Dioscorea hemsleyi* obtained using different extraction techniques inhibited both α-glycosidase (IC_{50} 27.41–274.36 μg/mL) and α-amylase (IC_{50} 3.66–47.57 μg/mL); however, the quantities required for α-amylase inhibition were much higher than those required for α-glycosidase. In contrast, Li et al. (2017) found that the *Sargassum pallidum* PSs showed weaker inhibitory effects on α-glycosidase than α-amylase. The α-glycosidase and α-amylase inhibitory activities may be related to the sugar and protein contents, monosaccharide composition, molecular weight, and chemical structures of HEPs. Therefore, on the basis of the above results, we conclude that the citric acid extraction is more useful in extracting HEPs from *H. erinaceus* with prominent antioxidant and hypoglycemic activities than the other extraction solvents.

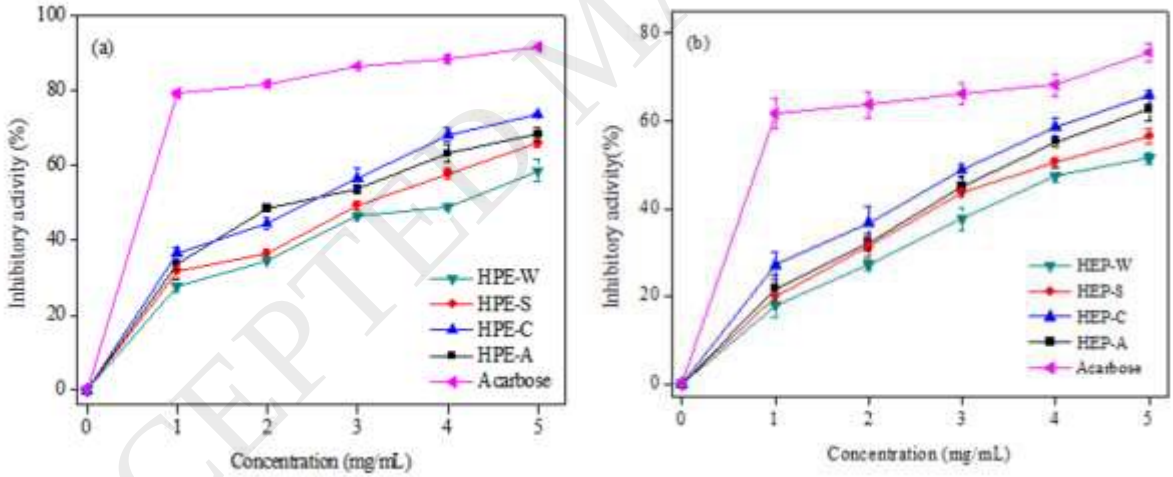

Fig. 7. Inhibitory effects of HPE-W, HPE-S, HPE-C, and HPE-A on the (a) α-glycosidase and (b) α-amylase activity. Each value is expressed as mean ± SD (n =3).

4. Conclusions

In summary, hot water, 0.9% NaCl, citric acid, and 1.25 M NaOH/0.05% $NaBH_4$ were

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used for the extraction of water-soluble PSs (HEP-W, HEP-S, HEP-C, and HEP-A) from the fruit body of *H. erinaceus*. The results showed that the extraction solvents had significant effects on the extraction yields, physicochemical properties, preliminary structural characteristics, and microstructures of the HEPs. HEP-C extracted with citric acid showed stronger antioxidant and α-glycosidase and α-amylase inhibitory activities in vitro than HEP-W and HEP-S and had a lower molecular weight. Therefore, citric acid as a safe food additive that can be widely used in food processing and for the extraction of bioactive PS. Further investigations on the isolation, purification, and structural characterization of HEP-C are ongoing.

Acknowledgements

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